IT 50700

Measurement and Evaluation in Industry and Technology

Term Paper Guidelines

Fall 2026

1 Introduction

You will be working in a group on a topic related to statistics and its applications. The instructor creates the groups randomly. Your group is to find a topic and to formulate a problem statement. You will submit a term paper report, a power point presentation, and individual peer evaluation in Brightspace by the due dates mentioned in the course syllabus.

2 The Goals

The goals are for students to:

- Apply statistics in a real-world situation of their choice.
- Be able to clearly report and present their findings.

Note 1: The presentation should be about 10 minutes. All group members must be available to present on one of the two dates mentioned in the course syllabus.

Note 2: all students must be present on both presentation days.

3 The Topic

Groups are to pick up a topic of their choice and report it to the instructor by the due date mentioned in the course syllabus. For this project, you are going to use either of the following databases: The U.S. Government's Open Data: www.data.gov, or Center for Disease Control and Prevention: www.cdc.gov.

Note: The project must involve hypothesis testing and goodness-of fit test. In addition, the study must include either ANOVA or Linear Regression.

4 Paper Structure

The paper should be clear and detailed enough for anyone to understand with appropriate tables, graphs, calculations, etc. In particular, you should:

- Provide a background on the problem and the rationale behind choosing it.
- Provide statement of the problem, hypotheses, methods used, discussion with graphs, equations, etc.
- Discuss literature review on the method you use (i.e., applications, etc.)
- Use at least, six references, three of which must be professional academic journal publications. The rest can be books, professional society publications, technical reports, etc. In addition, other sources, such as news outlets,magazines, etc. may be used. The required citation and reference format is IEEE.
- Include an appendix (if needed.)

5 The Report Format

The paper should be prepared according to the following guidelines:

- The term paper should be between 8 to 10 (full) pages, excluding the cover page, the executive summary, the table of contents, appendix (if any), and the reference.
- It should be double-spaced, Times New Roman typeface, 12-point font. The page margin should be 1 inch on all sides. The page number should be on the top right corner ("Page # of #").
- All tables, graphs, etc., should be numbered and referred to by their corresponding numbers.
- The document should have a cover page, and a table of contents, and a reference list.
  - The cover page should include the course title and number, the project title, the group members' names, and the date.
  - The reference page should be on a separate page.
- The report should be written in third-person language.
- IEEE style should be used throughout.

Note: You may use the template in Brightspace for both the report and the presentation.

6 Submission

Submit a PDF file of the report, the power file of your presentation, and the peer evaluation form in Brightspace by the due dates mentioned in the course syllabus.

Note 1: The term paper report and the power point file are group submission (i.e., one submission per group), while the status report and peer evaluation is individual submission.

Note 2: You must attend all the presentations to receive the credit for your term paper (see the grading rubrics).

Note 3: Once you submit all these documents, you may not modify them and/or provide another version.

7 Grading Rubrics

Requirement			Percent (%)
Status report			10
	Introduction	5	
	Literature review	5	
Term paper report	Methodology	10	30
	Discussion & conclusion	5	
	Report structure, citation, reference	5	
Term paper presentation			40
Peer evaluation			10
Presentation attendance			10
Total			100